

Untargeted metabolomics reveals microbial disruption, host tissue damage, and route-dependent vaccine protection during *Histomonas meleagridis* infection

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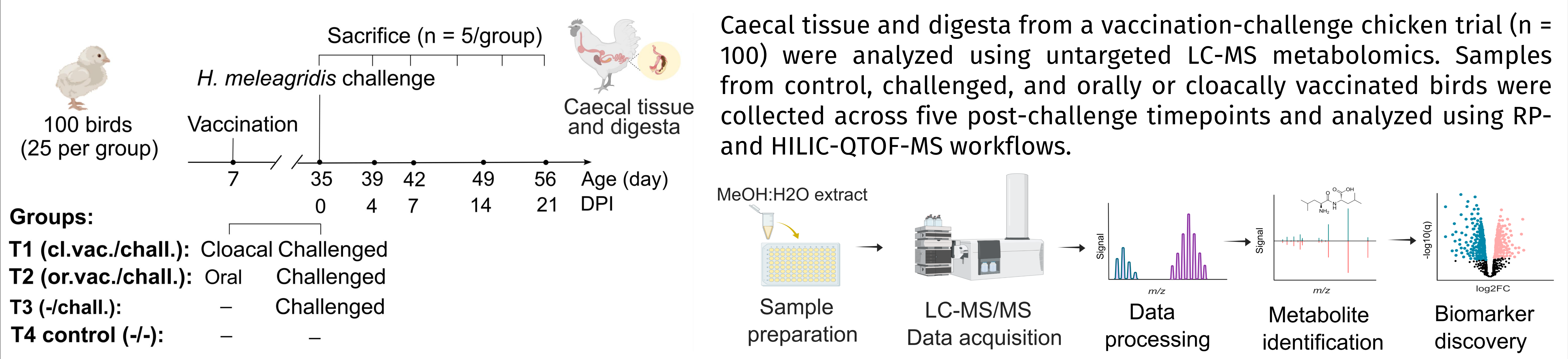
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Introduction

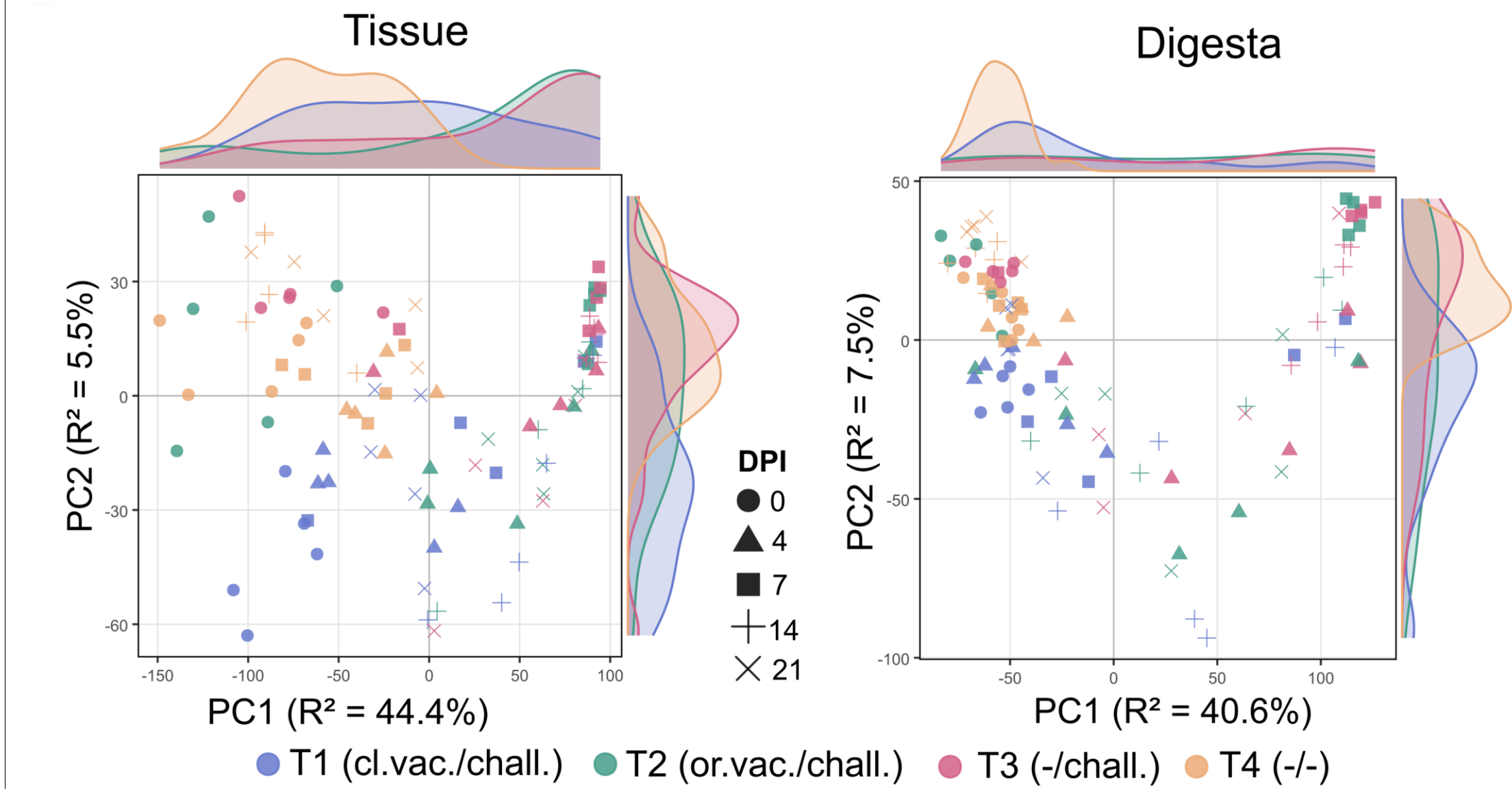
Host-microbe interactions are increasingly recognized as key determinants of immune function, infection outcomes, and vaccination responses. *Histomonas meleagridis*, an emerging enteric pathogen of poultry, causes severe caecal and liver lesions and substantial economic losses, yet the metabolic mechanisms underlying host-microbe interactions during infection remain poorly understood. Here, we used untargeted LC-MS metabolomics to characterize caecal metabolic responses to infection and evaluate the impact of different vaccination routes.

Experimental Design

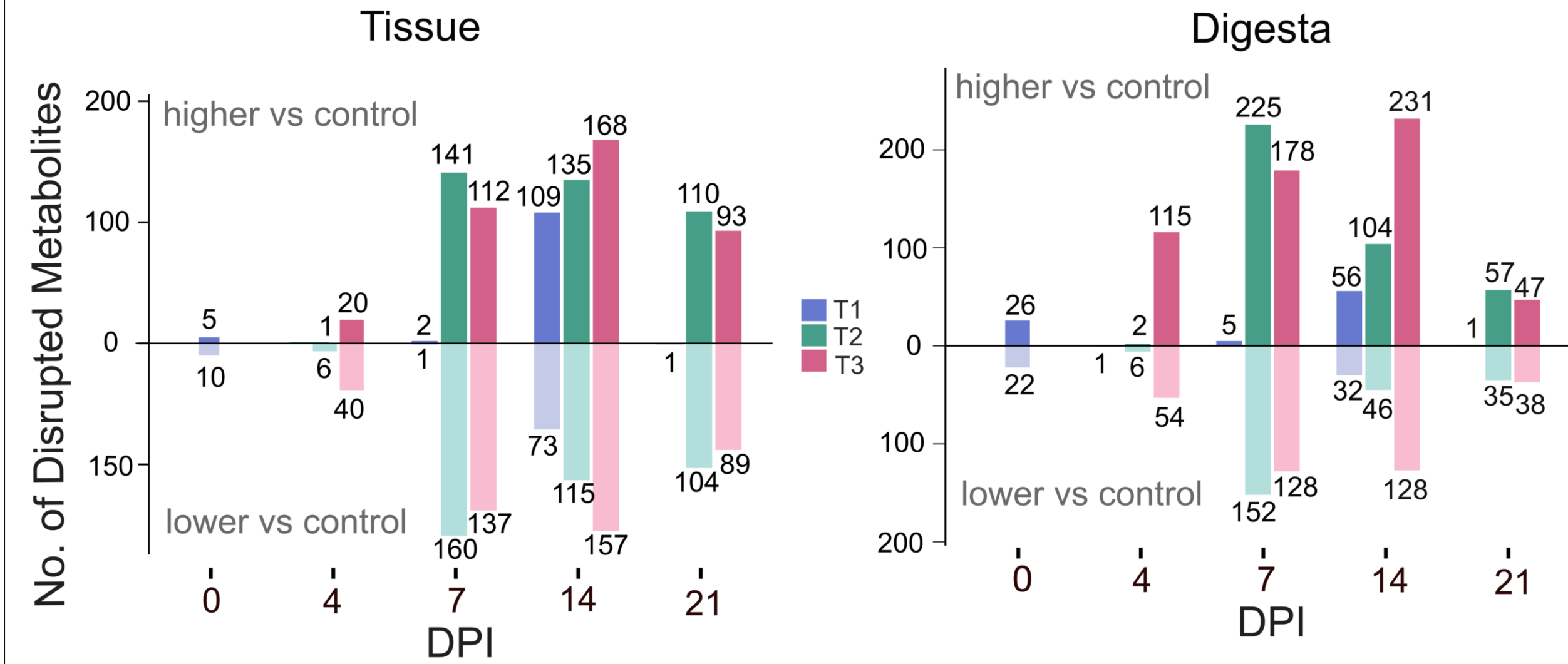


Results

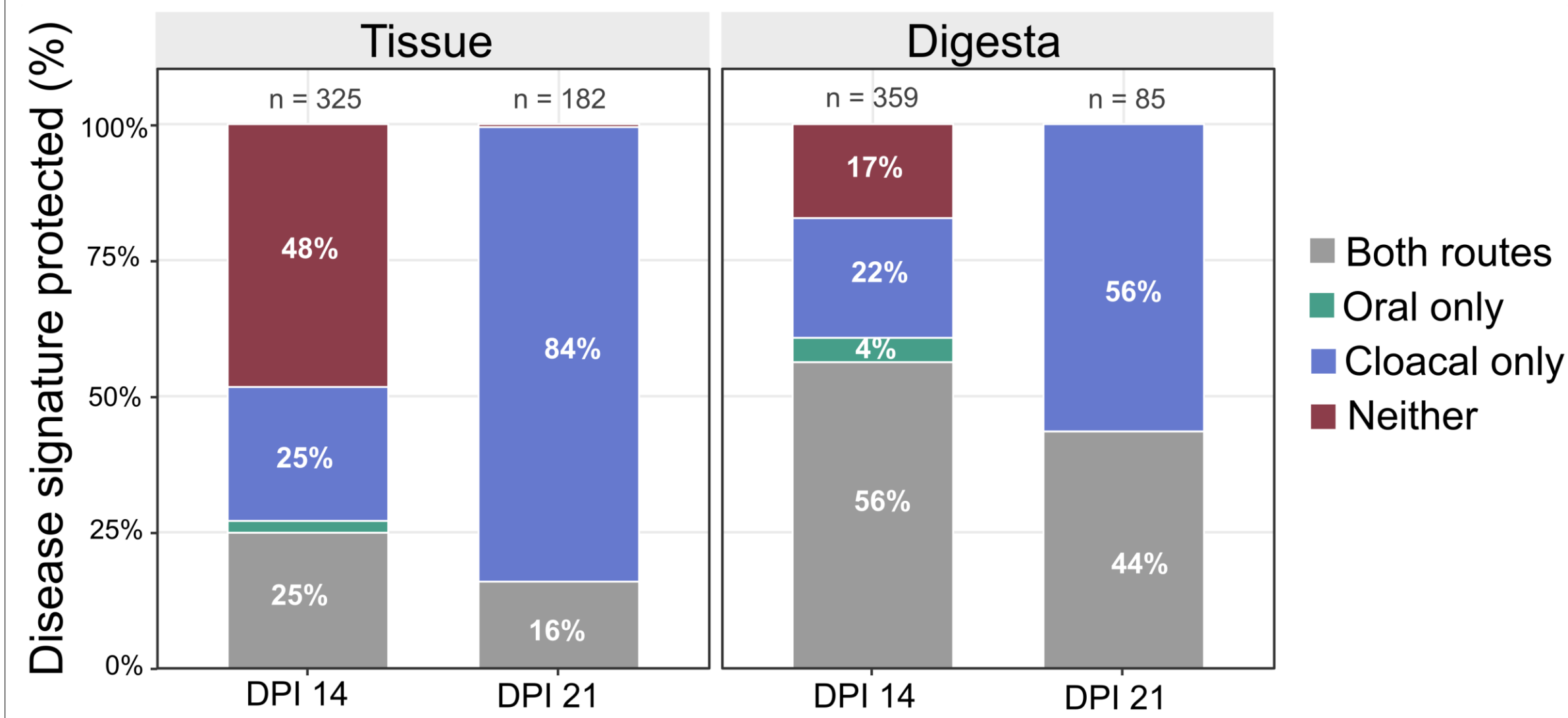
Infection broadly reshapes caecal metabolome



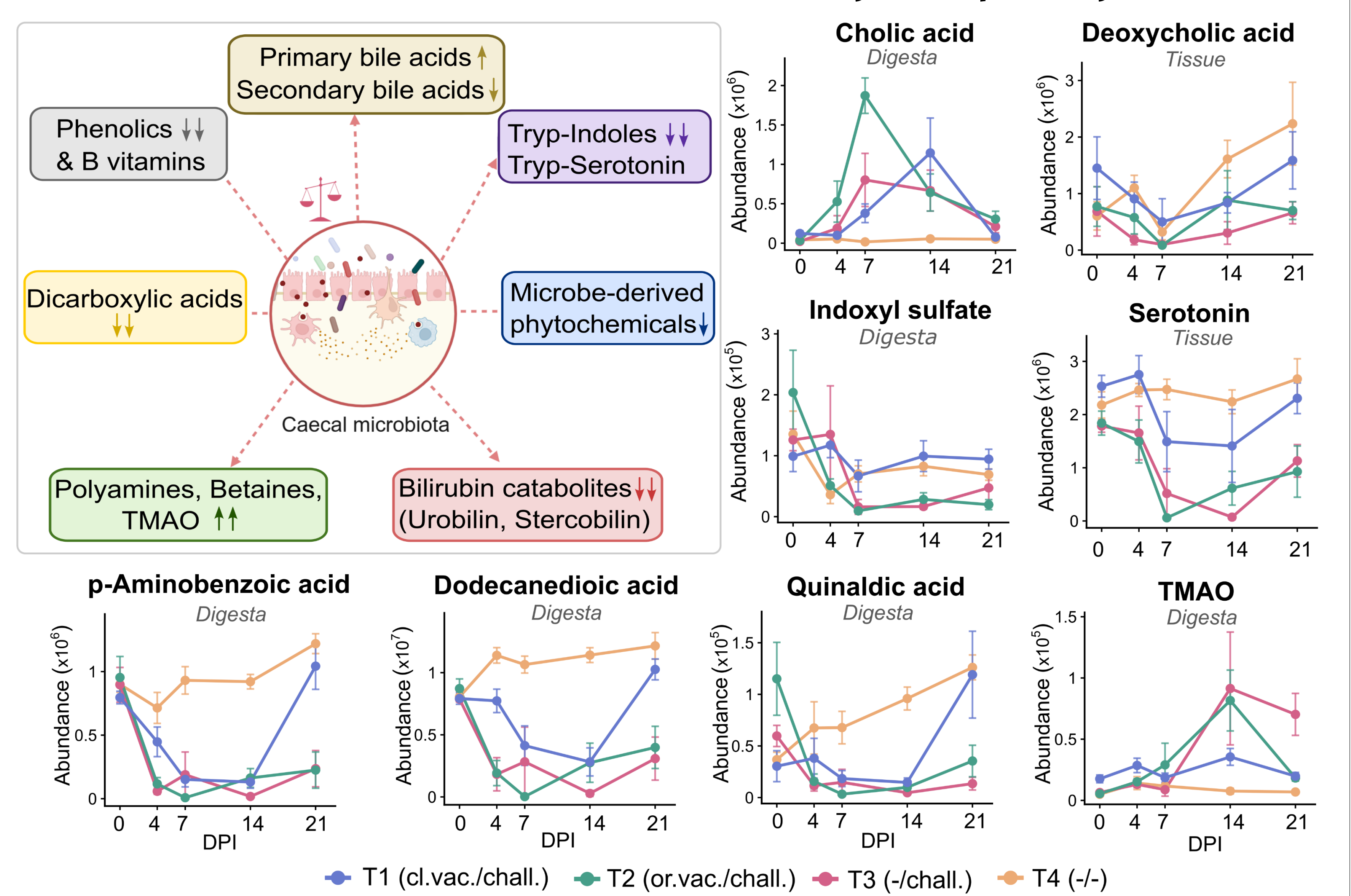
Temporal dynamics of metabolic disruption



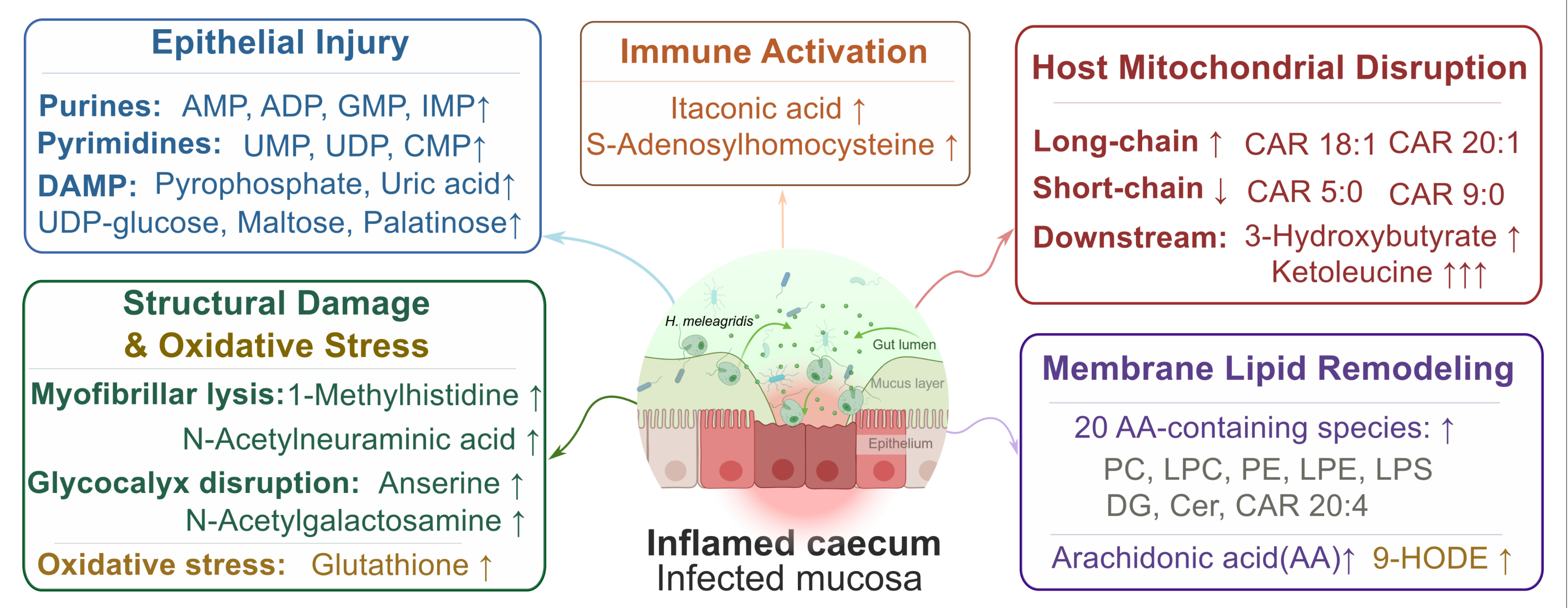
Vaccination preserves metabolic homeostasis



Microbiota-associated metabolites are broadly disrupted by infection



Infection causes tissue injury & metabolic reprogramming in caecum



Conclusions

H. meleagridis infection profoundly reshaped both the luminal and tissue metabolome, revealing extensive disruption of host-microbe metabolic crosstalk and loss of microbiota-associated functions. Cloacal vaccination largely preserved metabolic homeostasis and promoted faster recovery than oral vaccination. These findings provide the first caecum-focused metabolomic characterization of histomonosis and suggest that preservation of host-microbe metabolic interactions may be a key mechanism of vaccine protection.